

Can Epidemic Mechanics Guide Graph Networks? SEIR-Informed Spatio-Temporal Forecasting of Weekly Dengue in Sri Lanka

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Abstract—Graph neural networks can share information between neighbouring districts, but they know nothing about how a dengue infection progresses. We propose an SEIR-informed spatio-temporal graph network that passes a learned district graph into a decoder running a susceptible–exposed–infectious–recovered (SEIR) simulator. The decoder’s force of infection is scaled by district population and susceptible fraction, its initial exposed fraction is learned, and a learned gate blends its forecast with last week’s count. We evaluate the model on a rebuilt series of 559 weekly reports for 25 Sri Lankan districts, three weeks ahead. Rebuilding the series exposed a formula error in one published source table; on the earlier benchmark array, that single week moves the error of the persistence baseline from 44.80 to 29.52 cases per district-week. Across nine rolling forecast origins, the proposed model has the lowest test error of 30 compared models, 27.45 against 28.54 for persistence (paired difference -1.10, 95% interval -2.84 to 0.65). Scaling the force of infection by population lowers the error of a gated SEIR head from 28.43 to 27.62 on the same graph encoder. No difference from persistence is significant after adjustment for multiple comparisons, and a graph network with a residual head reaches 28.12. The mechanistic decoder therefore makes epidemic structure usable inside a graph network, but nine origins cannot yet show that it beats a strong simple baseline.

Index Terms—dengue forecasting, epidemic modeling, graph neural networks, persistence baseline, physics-informed learning, SEIR model

I. INTRODUCTION

Dengue causes recurring epidemics in Sri Lanka [1]. Forecasts of weekly cases for each district, a few weeks ahead, could help health officers plan clinical capacity and vector control. Spatio-temporal graph neural networks are a natural choice, because neighbouring districts exchange infections and share reporting patterns [2]. They learn only from past counts, however, and nothing in them encodes the incubation and recovery that shape an epidemic curve.

Compartmental models carry exactly that knowledge. An SEIR model tracks how people move from susceptible to exposed, infectious and recovered, and Liu et al. [3] showed that an LSTM can learn its force of infection for Sri Lankan dengue. We ask whether the same mechanism helps a graph network: does an SEIR decoder on top of a graph encoder

forecast weekly district counts better than the encoder alone, and better than repeating last week’s count? Persistence is the reference that matters most, because weekly counts change slowly; the benchmark this work builds on does not report it [2].

Our contributions are:

- 1) An SEIR-informed graph network: an adaptive graph encoder, a gated SEIR decoder whose force of infection is anchored to district population and susceptible fraction, and a learned initial exposed fraction.
- 2) An evaluation on a rebuilt and source-checked district series with nine rolling forecast origins, paired tests, and intervals against persistence, a graph baseline, and published spatio-temporal encoders.
- 3) Ablations that show what the mechanistic decoder contributes (free versus anchored rate, bare versus gated head, mass-action variant) and a measurement of how a source-table error changes the baseline.

II. RELATED WORK

Weng et al. [2] adapt graph encoders for Sri Lankan dengue forecasting and compare STGAT, A3T-GCN, ASTGCN, DCRNN, and AAGCN, which use graph attention, temporal attention, spatial-temporal attention, diffusion recurrence, and adaptive graph convolution [4]–[8]. Our STGAT follows their dengue adaptation, not the original pedestrian-trajectory model. DengueGNN also studies graph-based dengue prediction [9], and Graph WaveNet supplies the learned-adjacency construction of our adaptive encoder [10]. Liu et al. [3] learn a force of infection for a human SEIR model of Sri Lankan dengue with an LSTM. EINNs and CausalGNN place epidemiological structure inside neural forecasters, and MepoGNN couples graph learning with metapopulation dynamics [11]–[13]. We combine the two lines: a graph encoder with an SEIR decoder, tested against persistence on the same district series.

III. DATA

The rebuilt series contains 559 weekly reports for 25 districts, from June 15, 2013, to February 24, 2024; seven

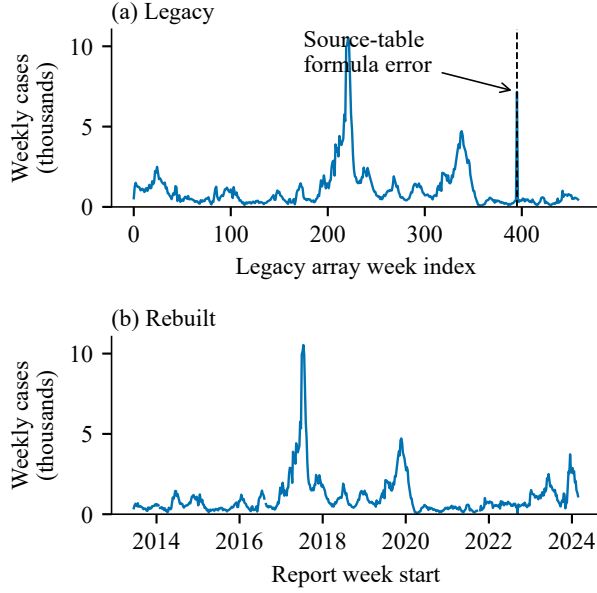


Fig. 1. National weekly cases (sum over 25 districts). (a) Legacy benchmark array, by row; the marked row comes from a published table with a formula error. (b) Rebuilt series used in this study; gaps are missing reports.

reports are missing and are never filled. Any forecast window whose inputs or targets touch a missing report is dropped. The task maps three past weeks of counts to the next three weeks. The earlier benchmark array, which we call the legacy array, has 459 rows and no verified calendar.

While rebuilding the series we checked every district row against the printed national total. Legacy row 395 comes from the report for December 26, 2020, to January 1, 2021 [14]. In its weekly row, 15 of 24 inner cells equal the sum of the two cells before them, and the districts sum to 7165 while the printed national total is 35. The same table’s year-to-date row, which equals the weekly count in the first week of a year, sums with the Kalmunai division to the printed total of 351. The spike in Fig. 1a is therefore a spreadsheet error, not an outbreak, and the rebuilt series uses the year-to-date row.

Districts are nodes of a graph whose edges are shared borders. Checking the border list against GADM polygons gives 57 shared borders, 114 directed entries, and 139 entries with self-loops. The graph used in the experiments below came from the benchmark and lists two further one-directional entries (Kandy to Ampara, Kegalle to Kalutara) that are not borders, 141 entries in total.

IV. SEIR-INFORMED GRAPH NETWORK

Fig. 2 shows the model. A graph encoder turns the input window into a representation per district, a read-out gives one value r per district and horizon, and a head turns r into a forecast, either directly or through an SEIR simulator.

A. Encoder and heads

Counts $c_{i,t}$ for district i and week t are standardized with training-period statistics,

$$z_{i,t} = \{\log(1 + c_{i,t}) - \mu\} / \sigma. \quad (1)$$

The persistence anchor p repeats the standardized latest count for every horizon. The GCN encoder projects the input to 64 dimensions and applies two residual graph layers, $H' = H + \text{Dropout}(\text{ReLU}(AHW + b))$, with A the row-normalized border adjacency with self-loops. The adaptive encoder learns $A_{\text{ad}} = \text{softmax}(\text{ReLU}(E_1 E_2^T))$ from standard-normal 25×10 node embeddings shared by both layers and exempt from weight decay. Published encoders can replace this stage. The direct, residual, and gated non-SEIR heads output r , $p + r$, and $p + g(r - p)$, with gate $g = \text{sigmoid}(a)$ and a initialized at -2 . The bare SEIR head outputs the standardized simulator count $z(\hat{c})$, and the gated SEIR head outputs $p + g\{z(\hat{c}) - p\}$. Training minimizes squared error on the standardized log scale; its minimizer is the conditional mean of $\log(1 + c)$, which, transformed back, is not the count-scale mean that RMSE rewards.

B. SEIR decoder

The decoder advances the susceptible, exposed, infectious, and recovered fractions of each district one day at a time with exponential flows,

$$\begin{aligned} f_S &= S(1 - e^{-\lambda \Delta t}), & f_E &= E(1 - e^{-\omega \Delta t}), \\ f_I &= I(1 - e^{-\gamma \Delta t}), & \hat{c}_{i,h} &= \rho_{\text{eff}} N_i \sum_{d=1}^7 f_E^{(d)}. \end{aligned} \quad (2)$$

Flows are computed from the current state and moved along $S \rightarrow E \rightarrow I \rightarrow R$; the weekly count is the sum of daily exposed-to-infectious flows, times the district population N_i and an effective reporting fraction. The rates are $\omega = 0.7/7 = 0.1$ and $\gamma = 1/7$ per day with seven daily steps, mean exposed and infectious periods of ten and seven days, following Liu et al. [3]. The base reporting fraction $\rho = 1/11$ follows the same work, and a learned scalar q gives $\rho_{\text{eff}} = \rho \exp(q)$. Initial infectious and exposed fractions come from $c_{t-1}/(\rho N_i)$ and $c_{t-2}/(\rho N_i)$, clipped to $[10^{-6}, 0.5]$. The susceptible fraction is $\text{clip}(0.318 - C_{<t}/(\rho N_i), 0.01, 1)$, where $C_{<t}$ is the sum of earlier reports. The value 0.318 is one minus the 68.2% seroprevalence measured in suburban Colombo [15], applied to every district. A learned multiplier $2 \text{sigmoid}(v)$ rescales the exposed fraction, either globally or per district from the encoder (learned E_0).

The force of infection can be free or anchored. A free rate exponentiates the read-out plus a learned bias, so the network must learn on its own the scale set by population and susceptibility. The anchored rate supplies that scale,

$$\lambda_{i,h}^* = \frac{\max(c_{i,t-1}, 0.5)}{7 \rho_{\text{eff}} N_i S_i} \exp\{\text{clip}(r_{i,h}, -10, 10)\}. \quad (3)$$

The first factor is the daily rate whose infections, reported at ρ_{eff} , reproduce last week’s count, so the read-out only

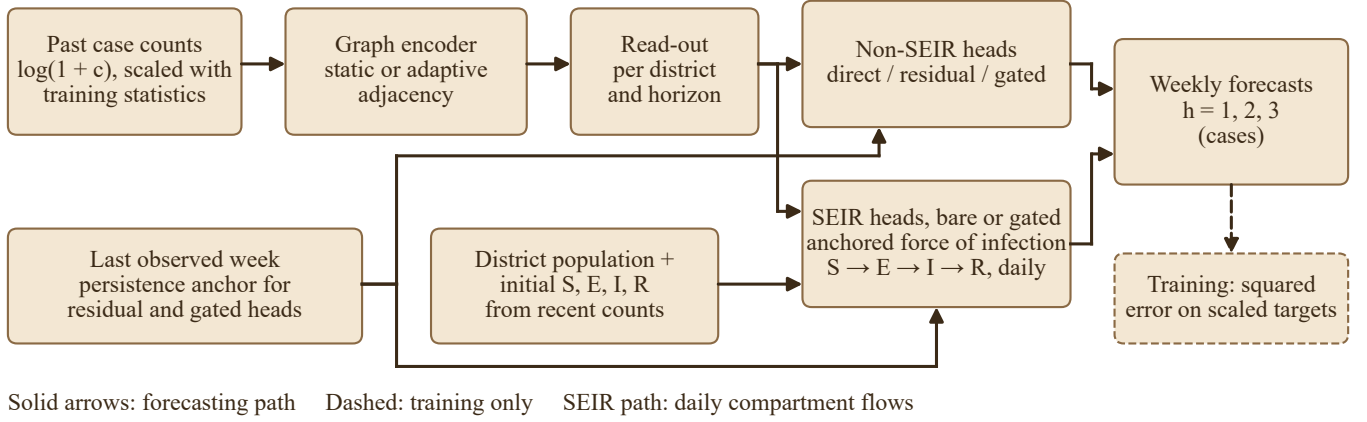


Fig. 2. SEIR-informed graph network. The encoder and read-out feed either a non-SEIR head or an SEIR head that converts a force of infection into weekly incidence through daily compartment flows. The last observed week is the anchor of the residual and gated heads; district population and initial compartments enter the SEIR head. The dashed path is used only in training.

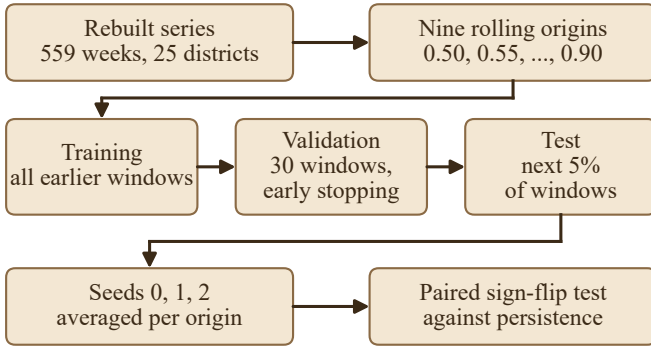


Fig. 3. Evaluation protocol. At each of nine origins the model is trained on earlier windows, stopped early on 30 validation windows, and tested on the next 5% of windows; seed-averaged origin scores are compared with persistence.

has to learn a departure from it. Both rates are multiplied by a learned spatial import term $\exp\{0.1\beta(u_i - \bar{u})\}$, with $u_i = \log \max((AI)_i, 10^{-9})$ from the border graph and $\bar{u}$ its mean over districts. A mass-action variant instead multiplies a learned daily transmission coefficient by the infectious fraction, recomputed each day, without the import term. We call the adaptive encoder with the gated SEIR head, anchored rate, and learned E_0 the Adaptive SEIR-GNN.

V. EXPERIMENTAL SETUP

We use rolling-origin evaluation (Fig. 3): the cut-off moves through the series, and the model is retrained at each origin. The nine origins lie at 0.50 to 0.90 of the forecast-start list in steps of 0.05, and each test block covers the next 5%. The 30 starts immediately before each cut-off form the validation set. We fixed this design after development runs with three origins (0.55, 0.70, 0.85; 15% test blocks) and a wider nine-origin grid (0.40 + $k/15$). Three origins cannot give an exact sign-flip p below 0.25, which motivated nine. Each learned model

is trained with seeds 0, 1, and 2, giving 27 runs per model; persistence is deterministic.

All models share one setting: Adam with learning rate 0.003 and weight decay 0.0001, batch size 32, cosine decay, gradient clipping at 5, up to 300 epochs, and early stopping on count-scale validation RMSE with patience 40. Because every window forecasts three weeks, the last training and first validation targets share two weeks, as do the last validation and first test targets. We compare 30 learned models with persistence: GCN, LSTM, STGAT, A3T-GCN, ASTGCN, and AAGCN, each with direct, residual, and free-rate gated and bare SEIR heads, and six adaptive-encoder and anchored variants.

Seeds are averaged within each origin, and each model is compared with persistence on the nine origin-level differences. We report RMSE and MAE in cases per district-week, RMSE by horizon, the paired difference with a 95% t interval, and the exact two-sided sign-flip test with Holm adjustment over the 30 models. With nine origins the smallest attainable adjusted p is 0.117, so the test can detect only differences that hold at every origin by a wide margin.

VI. RESULTS

A. Comparison with persistence and baselines

Table I gives the test results. Persistence has RMSE 28.54 and MAE 13.94; the GCN with a residual head, our baseline graph network, has 28.12 and 13.62. The Adaptive SEIR-GNN has the lowest test RMSE of all models, 27.45, a paired difference from persistence of -1.10 with interval [-2.84, 0.65]. Its raw sign-flip p is 0.199 and its Holm-adjusted p is 1.000, so the improvement is not significant. No learned model differs significantly from persistence after adjustment, in either direction. Fig. 5 shows that the intervals of the strongest models all lie mostly below zero but include it. Fig. 4 shows why: the Adaptive SEIR-GNN is below persistence at 7 of the 9 origins, but the origins differ far more from each other than the models do.

TABLE I

TEST RESULTS ON THE REBUILT SERIES (NINE ROLLING ORIGINS, THREE SEEDS). RMSE AND MAE ARE IN CASES PER DISTRICT-WEEK, AVERAGED OVER ORIGINS AND SEEDS; HORIZON COLUMNS REPORT RMSE. Δ IS THE SEED-AVERAGED PAIRED RMSE DIFFERENCE FROM PERSISTENCE WITH A 95% T INTERVAL OVER ORIGINS (NEGATIVE IS BETTER). p_{adj} IS THE EXACT SIGN-FLIP TEST WITH HOLM ADJUSTMENT OVER ALL THIRTY LEARNED ARMS. BOLD MARKS THE LOWEST TEST RMSE (THE PROPOSED ADAPTIVE SEIR-GNN); NO DIFFERENCE IS SIGNIFICANT AFTER ADJUSTMENT.

Model	$h = 1$	$h = 2$	$h = 3$	RMSE	MAE	Δ [95% CI]; p_{adj}
Persistence (last week)	23.47	27.94	33.26	28.54	13.94	—
GCN, direct	23.10	27.82	32.68	28.17	13.66	-0.37 [-1.88, +1.14]; 1.000
GCN, residual (baseline GNN)	23.24	27.79	32.47	28.12	13.62	-0.42 [-1.54, +0.70]; 1.000
LSTM, direct	23.65	28.22	33.38	28.73	14.00	+0.19 [-1.99, +2.37]; 1.000
ASTGCN, direct	23.92	28.54	33.45	28.94	14.27	+0.40 [-2.02, +2.81]; 1.000
AAGCN, direct	24.11	29.10	33.66	29.26	14.06	+0.72 [-1.57, +3.00]; 1.000
STGAT, direct	42.60	44.55	46.75	44.71	22.17	+16.17 [+4.24, +28.11]; 0.227
ASTGCN, residual	23.07	28.08	33.66	28.65	13.98	+0.11 [-1.68, +1.90]; 1.000
Adaptive GCN, residual	23.07	27.78	32.15	27.94	13.61	-0.60 [-1.81, +0.61]; 1.000
GCN, SEIR, free rate	27.58	40.81	50.14	40.77	17.97	+12.23 [+2.67, +21.80]; 0.227
GCN, gated SEIR, free rate	22.86	28.04	33.29	28.43	13.51	-0.11 [-3.23, +3.00]; 1.000
Adaptive GCN, SEIR, anchored	23.99	28.81	35.29	29.80	13.84	+1.26 [-1.47, +3.98]; 1.000
Adaptive GCN, gated SEIR, mass-action	22.90	28.02	33.19	28.39	13.52	-0.16 [-3.16, +2.85]; 1.000
GCN, gated SEIR, anchored	22.49	27.38	32.04	27.62	13.39	-0.92 [-2.60, +0.76]; 1.000
Adaptive GCN, gated SEIR, anchored	22.51	27.09	31.85	27.45	13.31	-1.09 [-2.81, +0.63]; 1.000
Adaptive GCN, gated SEIR, anchored, learned E_0	22.45	27.07	31.88	27.45	13.29	-1.10 [-2.84, +0.65]; 1.000

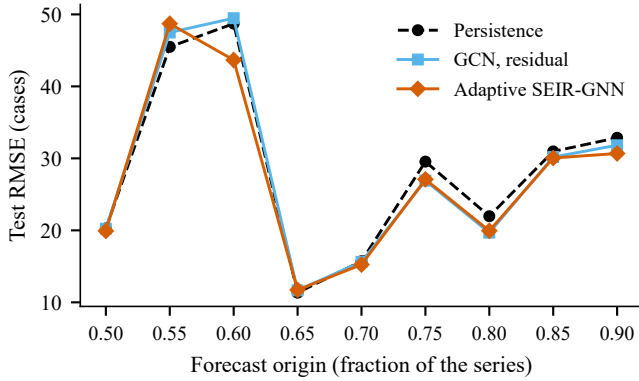


Fig. 4. Seed-averaged test RMSE at each of the nine forecast origins.

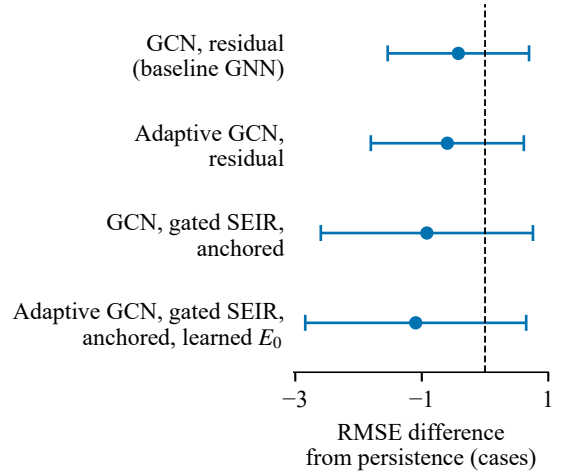


Fig. 5. Paired test RMSE differences from persistence with 95% t intervals over nine seed-averaged origins. Negative values favour the model.

B. What the SEIR decoder contributes

Fig. 6b compares the mechanistic heads. On the GCN encoder, anchoring the force of infection lowers the gated SEIR head from 28.43 to 27.62. Gating matters as much as anchoring: the bare anchored SEIR head on the adaptive encoder reaches 29.80, above persistence. The mass-action variant reaches 28.39, and the anchored gated head scores 27.45 with or without the learned E_0 . The adaptive encoder with a residual head and no SEIR decoder scores 27.94. That comparison changes the head and the anchor together, so it does not isolate the SEIR equations. On the three-origin protocol, a gated head without the simulator is within 0.68 of the gated SEIR head on validation for STGAT, A3T-GCN, and DCRNN, while the SEIR version has lower test error for all three.

C. Horizon and data quality

The Adaptive SEIR-GNN is below persistence at every horizon, by 1.03, 0.86, and 1.38 cases (Fig. 6a). Persistence rises from 23.47 at one week to 33.26 at three weeks; the GCN residual baseline goes from 23.24 to 32.47. On the legacy array, removing the six windows that touch the erroneous row lowers mean persistence RMSE from 44.80 to 29.52, and the last fold alone from 68.62 to 22.80 (Fig. 6c). One spreadsheet error thus changes how hard the task appears by more than a third, while the best model differs from persistence by 1.10.

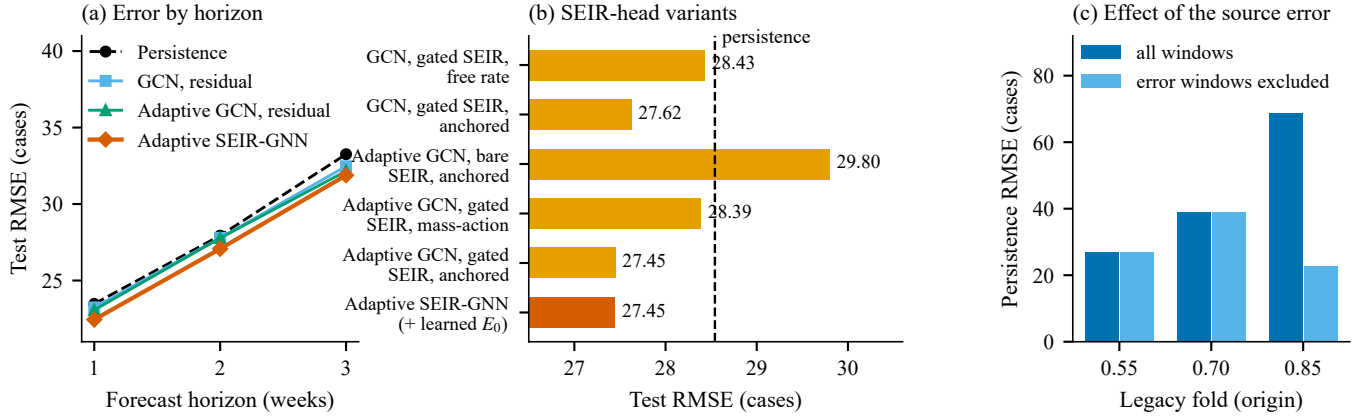


Fig. 6. (a) Test RMSE by forecast horizon. (b) Test RMSE of the SEIR-head variants; the dashed line is persistence. (c) Persistence RMSE on the three folds of the legacy array, with all windows and without the six windows that touch the erroneous source row. Panels (a) and (b) use the nine-origin protocol.

VII. DISCUSSION AND LIMITATIONS

The SEIR decoder becomes useful once the force of infection carries the population and susceptible scale and a gate lets the model fall back on last week's count. Without the anchor, the head must learn a district-specific scale it never sees; without the gate, a bare simulator overshoots or undershoots and loses to persistence. We believe this is why the anchored, gated decoder gives the lowest errors, but the experiments do not separate the contribution of the SEIR equations from that of the gate and anchor. Persistence is likely hard to beat because weekly counts change slowly over three weeks, while a learned model must estimate its parameters from few epidemic regimes.

Several limitations apply. The nine origins were chosen after development runs on other origin designs, and the six adaptive and anchored variants were added after earlier results were seen, so these results are development evidence, not confirmation on unseen weeks. Two target weeks overlap at each partition boundary, which lets early stopping see some outcomes of the first test windows. The graph used in the experiments has two entries that are not shared borders. All models share one training setting rather than an equal tuning budget, so the weaker published encoders may be under-tuned. The fixed rates, the reporting fraction, and the Colombo seroprevalence are assumptions, and the decoder has no serotypes, waning immunity, or mosquito compartment. The study covers one country, one disease, and a three-week horizon. Our next step is an evaluation on weeks after February 2024 with the corrected graph, a matched gated head without SEIR, equal tuning, and seasonal and autoregressive baselines, all fixed in advance.

VIII. CONCLUSION

We proposed an SEIR-informed graph network for weekly district-level dengue forecasting in Sri Lanka. Anchoring the force of infection to population and susceptibility, and gating the simulator with last week's count, gives the lowest test error

among 30 models, 27.45 against 28.54 for persistence. The gain is not significant on nine origins, and a single source-table error changes the baseline far more than the best model's gain. Epidemic mechanics can be made to work inside a graph network; showing that they improve forecasts needs a longer, untouched test period.

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